

auto_model Function: Automated Model Comparison Tool

What the Function Does

The auto_model function takes any dataset and a target variable, then automatically fits five different machine learning models, evaluates them using 10-fold cross-validation, and returns ranked comparison tables. Instead of manually writing out the code for each model every time, this function does it all in one line.

The function fits these models:

1. Linear Regression
2. CART (decision tree)
3. Random Forest
4. Gradient Boosting (GBM)
5. Support Vector Machine (SVM)

For each model, it reports the CV RMSE, the test set RMSE, and the stability (standard deviation across folds). It also generates a comparison plot.

Testing on 2 Datasets

I tested the function on both datasets from the previous assignments to make sure it works and to show how different data structures lead to different results.

Carseats Dataset

Model	CV RMSE	CV RMSE SD	Test RMSE
Linear Regression	1.00	0.13	1.14
GBM	1.12	0.16	1.43
SVM	1.17	0.17	1.39
Random Forest	1.55	0.22	1.66
CART	2.20	0.32	2.04

Best model: Linear Regression, Most stable model: Linear Regression

Mall Dataset

Model	CV RMSE	CV RMSE SD	Test RMSE
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CART	21.77	3.56	21.23
SVM	22.22	3.52	22.05
GBM	22.87	3.53	21.00
Random Forest	23.13	5.00	17.65
Linear Regression	24.59	2.71	25.53

Best model: Random Forest (on test set), Most stable model: Linear Regression

What This Shows

The function correctly identifies that different datasets need different models. On Carseats, where the relationships are mostly linear, Linear Regression performs best. On Mall, where the relationships are nonlinear and complex, Random Forest performs best.

This is the main point of building a reusable tool like this. Instead of assuming one algorithm is always best, the function tests multiple approaches and lets the data decide which one works.

Why This Matters

Before this project, I was writing out the code for each model separately every time. That works, but it takes a lot of time and it's easy to make mistakes or forget to use the same CV setup across all models.

With `auto_model`, I can compare five models on any dataset with a single line of code. The function handles the train/test split, sets up cross-validation, trains all the models, and outputs clean comparison tables. This tool automates repetitive tasks so that it can focus on interpreting results instead of rewriting the same code over and over.

The next step would be to add more features to the function, like support for classification problems, more model options, or automatic ensemble building. But even in its current form, it saves a lot of time and makes model comparison much easier.

Plots:

